MCO-1D Mark 1 — As-Built Physical Measurement & Calibration Sheet

AUTHORITATIVE SHOP-FLOOR HANDOFF MANIFEST FOR FIRMWARE LAWS & KINEMATIC GOVERNORS (REV 2)

AIRFRAME S/N: MCO-1D-001	**BUILD DATE:** ____ / ____ / 2026	**SHOP LOCATION:** _____
AIRFOIL: [ ] S1223 [ ] SD7062	**AMBIENT TEMP:** _____ °C	**DIGITAL SCALE S/N:** _____
LEAD AVIONICS ENG: _____	**ATMOS PRESSURE:** _____ hPa	**CALIPER / RULE S/N:** _____

1. Airframe Mass, Wheel Reactions & Longitudinal Center of Gravity

Accurately weigh flight-ready airframe with 6S 2200mAh LiPo and all 8 wings installed. Firmware dynamically updates stall speed Vs, climb Ps, and hover T/W from these values.

Param ID	Calibration Parameter	Baseline / Nominal Target	As-Built Measured Value	Tolerance / Limits	Firmware Coupling & Action
M-01	**All-Up Flying AUW** **All 8 wings + 6S pack installed**	***3,642 g (S1223)*** ***3,636 g (SD7062)*** *(Sub-4 kg envelope)*	**[ ] g**	Max MTOW: **< 4,000 g** Target: 3,550–3,750 g	**PRIMARY GATE**: Recalculates stall speed floor Vs ∝ √(m), required lift L=mg, Boyd Ps, and hover T/W ratio (1.44:1 nominal).
M-02	**3-Point Landing Gear Tare** **Calibrated scales under wheels**	*Tailskid: ~10% (~360 g)* *Mains: ~45% ea (~1,640 g ea)*	**Tail: [ ] g** **Port: [ ] g** **Stbd: [ ] g**	Port vs Stbd balance within **± 20 g**	Confirms longitudinal ground stability against tail tipping and verifies transverse lateral symmetry across gear saddle.
M-03	**Longitudinal CG Station** **Spine knife-edge balance**	***352.6 mm aft datum*** *(Sled Hole 3: 320mm arm)*	**[ ] mm**	352.6 ± 5.0 mm (Safe: 338–385 mm)	Verifies static margin (+48.3 mm / +33.6% MAC forward of neutral point X_NP = 400.9 mm) for positive pitch stability.

2. Wing Root Collar Geometry & Longitudinal Cascade Stagger (Central Spine)

Measure collar bearing centers along the 700mm carbon spine from fuselage nose tip (Datum 0.0mm). These dimensions govern dynamic tip clashing, inter-wing sweep clearances, and wake slipstream capture.

Param ID	Geometric Feature	Nominal Design Target	As-Built Measured Value	Tolerance	Governing Physics & Impact
M-04	**Root Lateral Spacing** **(2 · Y_root across spine)**	***70.0 mm*** *bearing centers* *(Y = ±35.0 mm from spine)*	**[ ] mm**	70.0 ± 1.5 mm	**CRITICAL TIP CONTACT GEOMETRY**: Governs tip separation ΔY(θ) = 2(Y_root + (b/2)cosθ). Sets software stroke clamp θ_max ≤ 85°.
M-05	**Station 1 Arm (W1 Fore)** **Pivot center from nose datum**	***X1 = 95.0 mm*** *aft datum* *Waterline:* ***Z1 = +48.0 mm***	**X1: [ ] mm** **Z1: [ ] mm**	X: ± 1.5 mm Z: ± 1.0 mm	Leading lifting pair arm; governs nose pitching moment and canard dynamic upwash shed onto downstream stations.
M-06	**Station 2 Arm (W2 Mid-Fore)** **Pivot center from nose datum**	***X2 = 275.0 mm*** *(ΔX = 180mm)* *Waterline:* ***Z2 = +16.0 mm***	**X2: [ ] mm** **Z2: [ ] mm**	X: ± 1.5 mm Z: ± 1.0 mm	Second cascade station; establishes 90° metachronal wake capture timing τ = ΔX / V_inf for induced drag reduction.
M-07	**Station 3 Arm (W3 Mid-Aft)** **Pivot center from nose datum**	***X3 = 455.0 mm*** *(ΔX = 180mm)* *Waterline:* ***Z3 = -16.0 mm***	**X3: [ ] mm** **Z3: [ ] mm**	X: ± 1.5 mm Z: ± 1.0 mm	Third cascade station; vertical drop (-32mm step) prevents chord-plane clash during differential sweep maneuvers.
M-08	**Station 4 Arm (W4 Aft)** **Pivot center from nose datum**	***X4 = 635.0 mm*** *(ΔX = 180mm)* *Waterline:* ***Z4 = -48.0 mm***	**X4: [ ] mm** **Z4: [ ] mm**	X: ± 1.5 mm Z: ± 1.0 mm	Trailing station; downstream slipstream ejector. Governs pitch damping and overall tandem neutral point X_NP = 400.9 mm.
M-09	**Spar Quarter-Chord Offset** **Root rib LE to spar center**	***36.0 mm*** *aft of root LE* *(Strict 25.0% MAC target)*	**[ ] mm**	36.0 ± 0.5 mm (24.5–25.5% MAC)	**AERO HINGE MOMENT GATE**: Eliminates aerodynamic moment arms on spar pivot; prevents pitching blowback on B24CLM pitch servos.

3. Individual Wing Panel Matrix (Semi-Span, Mass, Neutral Pitch & Twist)

Measure each of the 8 assembled wing panels independently prior to airframe mounting. Ingested into ODrive FOC inertia matrices, roll trim schedules, and aerodynamic drag polar estimators.

Wing Station	Side	Semi-Span (bY2) (Root to Tip)	Panel Mass (m) (Covered Panel)	Neutral Pitch (α0) (@ 1500µs PWM)	Tip Washout (ε0) (Root vs Tip Twist)	Top Frequency (Dyneema Line)
W1 Fore	Port (L)	**[ ] mm**	**[ ] g**	**[ ] °**	**[ ] °**	**[ ] Hz**
W1 Fore	Stbd (R)	**[ ] mm**	**[ ] g**	**[ ] °**	**[ ] °**	**[ ] Hz**
W2 Mid-Fore	Port (L)	**[ ] mm**	**[ ] g**	**[ ] °**	**[ ] °**	**[ ] Hz**
W2 Mid-Fore	Stbd (R)	**[ ] mm**	**[ ] g**	**[ ] °**	**[ ] °**	**[ ] Hz**
W3 Mid-Aft	Port (L)	**[ ] mm**	**[ ] g**	**[ ] °**	**[ ] °**	**[ ] Hz**
W3 Mid-Aft	Stbd (R)	**[ ] mm**	**[ ] g**	**[ ] °**	**[ ] °**	**[ ] Hz**
W4 Aft	Port (L)	**[ ] mm**	**[ ] g**	**[ ] °**	**[ ] °**	**[ ] Hz**
W4 Aft	Stbd (R)	**[ ] mm**	**[ ] g**	**[ ] °**	**[ ] °**	**[ ] Hz**
NOMINAL TARGETS	Both	***1,150.0 mm*** *(±2.0)*	***69.0 g*** *(±2.5 g)*	***0.0°*** *(±0.5°)*	***0.0° to -1.5°***	***210 Hz*** *(160 N)*

4. Drivetrain, Tension, Kinematic Stops & Sensor Zero Calibrations

Param ID	Subsystem / Mechanism	Baseline Target	As-Built Measured Value	Tolerance	Firmware Safety Action
M-10	**Dyneema Rest Pre-Tension Acoustic tap pitch (150mm span)**	***160.0 N (210 Hz)*** *post-48h creep settlement*	**[ ] N Pitch: [ ] Hz**	160 ± 10 N (210 ± 8 Hz)	Eliminates slack-side cable derailment across reversals while protecting motor bearing radial life.
M-11	**Wingtip Dynamic Backlash Mechanical slop at 1.15m tip**	***< 1.0 mm play*** *at tip (< 0.05° angular slop)*	**[ ] mm**	Max: 1.5 mm Target: < 0.8 mm	Ensures precision aerodynamic angle tracking; eliminates aeroelastic flutter trigger at high speed.
M-12	**Sector Wheel Travel Arc Mechanical stop-to-stop arc**	***200.0° Total Arc*** *(±100.0° from neutral)*	**[ ] deg**	198.0° to 202.0° Stop sym: ± 1.0°	Sets physical stops at ±100°; flight governor clamps dynamic stroke to f·θ0 ≤ 91.7°·Hz (≤1800 RPM).
M-13	**Pitot-Static Zero Offset MS4525DO indoor zero reading**	***0.00 Pa*** *(covered tube) Calm room ambient*	**[ ] Pa**	± 2.0 Pa (± 0.02 hPa)	Calibrates low-speed rolling median filter before synthetic EKF3 airspeed blending takes over.

5. Mandatory Pre-Flight Hardware Assembly Signoff Checklist

[ ] 0. Sub-4 kg Flight AUW Verification	Weigh complete flight-ready airframe with 6S 2200mAh pack. Confirm < 4,000 g (nominal 3,642 g). Record M-01.
[ ] 1. Dyneema 48-Hour Creep Settlement	Cables rested at 200 N for 48h; re-tensioned to 160 N ± 10 N (210 Hz tap pitch); witness marks aligned; shrouds installed.
[ ] 2. Carbon Spar Internal Ferrules & Clamps	6061-T6 ferrules (Ø8.0x40mm) epoxied into spar roots; spar clamp bolts torqued to 2.2 N·m with Loctite 243.
[ ] 3. Motor Thermal Mounts & NACA Duct Cooling	0.5mm FR4/silicone isolator gaskets under all 8 GoolRC motors torqued to 1.8 N·m; canopy NACA ram-air scoop unobstructed.
[ ] 4. Center of Gravity Knife-Edge Balance	Battery tray sled pin locked into Hole 3 (320mm arm); airframe balances on knife-edge at 352.6 ± 5.0 mm (Record M-03).
[ ] 5. Dual-Zone BEC Bus Electrical Droop Test	Full 16-servo step deflection: Zone A & B rails hold ≥ 7.1V (droop ≤ 0.3V); 4,700µF buffer capacitors verified intact.
[ ] 6. Biomimetic Pterostigma Flutter Balance	1.5g brass slugs verified at y=0.85b, x/c≈0.08 in all 8 wings; chordwise CG forward of elastic axis (≤ 24% chord).
[ ] 7. CAN-FD Termination Bus Resistance	DMM measurement across CAN_H and CAN_L reads exactly 60.0 Ω ± 1.5 Ω with two 120-Ohm split termination resistors.
[ ] 8. Dynamic Kinematic Governor Floor Clamping	FCC confirms f·θ0 ≤ 91.7°·Hz universal governor clamp active; ODrive velocity trips prevented below 1,800 RPM ceiling.

6. Production Handoff & TC-01 Bench Authorization

HANDOFF GATE: Mark 1 Physical Calibration	**STATUS: [ ] READY FOR FIRMWARE INJECTION [ ] RE-WEIGH / ADJUST**
LEAD AIRFRAME BUILDER: _____	**SIGNATURE & DATE: _____**
AVIONICS & FIRMWARE LEAD: _____	**FIRMWARE COMMIT HASH: _____**
FLIGHT TEST DIRECTOR: _____	**TC-01 GROUND RUN APPROVAL: [ ] APPROVED [ ] HOLD**